TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Biophotonic Chip Integration and Biocompatible Packaging for Diagnostics and Sensing**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (X), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Bio		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Optical Application Devices	Other Optical Application Devices
Concept of Proposed Technology	Development of an integration and packaging platform to couple photonic biosensor chips with microfluidics, light sources, and detectors in a compact, biocompatible, and manufacturable format. The solution targets disposable cartridges and instrument interfaces enabling label-based (fluorescence) and label-free (interferometric/RI) measurements for point-of-need and laboratory diagnostics, as well as chemical/biological sensing. Core features include passive optical alignment, microfluidic sealing, biocompatible encapsulation, and system-level validation.			
Rationale for Joint R&D Project	As an innovation center, we focus on developing advanced enabling technologies that integrate seamlessly into partners' instruments and workflows. Joining this R&D project allows us to co-create a modular platform that strengthens Korean SME's products while accelerating technology transfer and real-world adoption.			
Key Words	1. Biophotonics 2. Lab on chip 3. Microfluidic integration 4. Diagnostic sensors			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 2	End of the Project	Level 5
R&D Objectives	• Define a modular biophotonic chip-to-world packaging stack (optical, fluidic, electrical, thermal). • Build a prototype diagnostic device including a biophotonic chip • Demonstrate application-relevant performance with Korean SME-provided samples/use cases.			

R&D Target Specifications	• Leak-tight fluidic system to guide test liquid to photonic bio chip
	• Modular approach offering following functions:
	  ○ Integrated fluid reservoirs, e.g., blisters or cartridges to hold reagents, washing solution etc.
	  ○ Filtering of test liquid
	  ○ In-line air bubble removal
	  ○ In-line heating
	• Bio-compatibility: materials/processes selected to pass ISO 10993-5 cytotoxicity
	• Manufacturability: cartridge cost target < 10 USD at scale (not including photonic chip)			

R&D Plan	• Requirements & architecture: consolidate use-cases, define requirements & specifications, select materials/processes. • Integration & packaging processes, microfluidic sealing, bio-compatible encapsulation. • Module prototypes: build demo modules with disposable cartridges. • Verification & reliability: function testing, bio-compatibility screening. Application validation & scale-up: usability, assembly yield improvements, pilot transfer plan.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	CSEM: system architecture; photonic/electro-optical and microfluidic co-design; packaging process development; assembly; material selection; prototyping; reliability testing; coordination and documentation.
Expected R&R of Korean SME Partner	Korean SME(s): define product requirements and specifications; supply samples/reagents; contribute to cartridge usability and instrument interface; perform application testing and verification; assess manufacturability and localization; prepare regulatory and market pathway.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Deliverables: packaging design rules, system design, validated biophotonic module prototypes with disposable cartridges, verification dataset, pilot transfer plan. Impact: accelerated path for Korean diagnostics/sensor SMEs to integrate photonic chips into products, reduced development risk and time-to-validation, and foundation for manufacturing.

6. References

1. EU BioFOS Project: ID 611528
2. EU PhotonMED Project: ID 101139777
3. EU BIOCDx Project: ID 732309
4. EU MedPhab Project: ID 871345
5. EU Photo-Sens Project: ID 965643

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization